

Factor-Based Risk Parity for US Public Pension Funds

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1. Introduction and Research Objective

This project develops a Macro Factor Risk Parity Framework tailored for large US Public Pension funds, and constructs and back-tests a portfolio according to this framework. The portfolio is benchmarked against three references: (1) a classic 60/40 portfolio, (2) a CalSTRS-style allocation based on the California State Teachers' Retirement System, and (3) the realized aggregate allocation of all US public pension plans, reported by the Center for Retirement Research's Public Plans Database (PPD, ~5,000 plans, AUM ~US\$5.4T as of FY2024).

The core objective of this project is to build a pension fund portfolio allocation such that the four key macro risk factors – Growth, Inflation, Real Rates and Credit Spreads – each contribute an equal share of the portfolio's total systematic risk.

2. Motivation: Why Asset-Class Diversification May Be Misleading

Traditional asset-class based diversification carries a structural vulnerability: distinct asset classes, across domestic and international equities, fixed income, real estate and alternatives, share the same underlying macro factor exposures.

When a macro shock strikes, as in 2022, when inflation surged, cross-asset correlations rapidly converge toward 1. The intrinsic driver of the severe drawdowns is not the portfolio's ostensibly diversified asset-class labels, but rather its hidden macro factor concentration.

3. Data and Portfolio Proxies

We use monthly data from January 2004 to January 2026. The asset universe includes U.S. equity, international equity, U.S. bonds, TIPS, real estate, and commodities. Bond and TIPS returns are constructed from yield changes using a duration-based return approximation. These proxies allow us to compare our portfolio strategy with three benchmarks: a traditional 60/40 portfolio, a CalSTRS-style portfolio, and an aggregate U.S. public pension allocation. To ensure comparability, each benchmark's original asset categories are mapped into our six proxy asset classes. Detailed data sources, transformations, and benchmark-to-proxy mappings are provided in the accompanying Excel file.

The macro factor set includes Growth, Inflation, Real Rates, and Credit Spreads. Growth and Inflation use 12-month changes in industrial production and CPI, while Real Rates and Credit Spreads use monthly changes in the 10-year TIPS yield and the Moody's Baa-Treasury spread. We map all portfolios into this common factor space to test whether asset-class diversification translates into true macro risk diversification.

4. Methodology

We estimate each asset class's exposure to Growth, Inflation, Real Rates, and Credit Spreads using monthly return regressions. The resulting factor loading matrix B maps asset-class portfolios into macro factor exposures:

$$r_{i,t} = \alpha_i + \beta_i F_t + \epsilon_{i,t}.$$

We then decompose asset-class covariance into systematic macro-factor risk and residual idiosyncratic risk:

$$\Sigma_a = B\Sigma_f B^\top + D$$

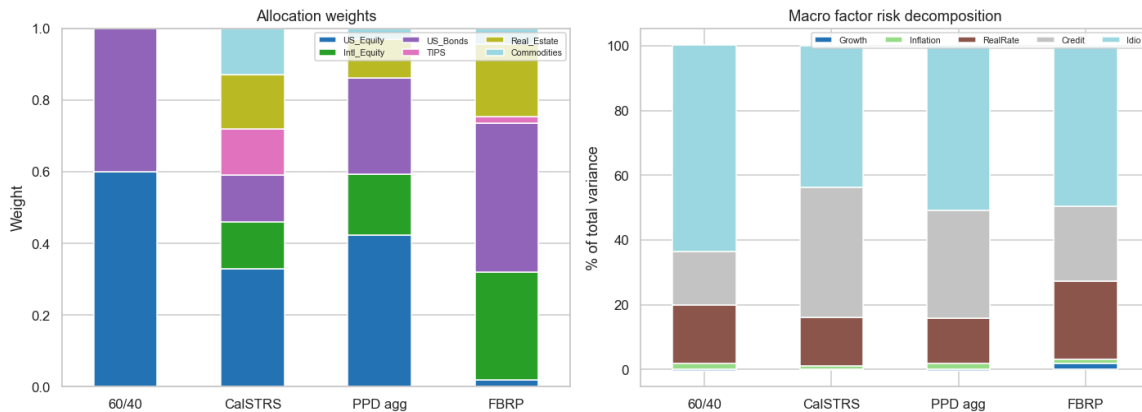
For each portfolio, we compute each macro factor's contribution to total systematic variance. The FBRP portfolio then chooses long-only weights that minimize the deviation of each factor's risk share from an equal-risk target, subject to realistic pension allocation bounds:

$$\min_w \sum_k (s_k(w) - 1/K)^2$$

This framework evaluates diversification by underlying macro risk contributions rather than by asset-class labels.

5. Results

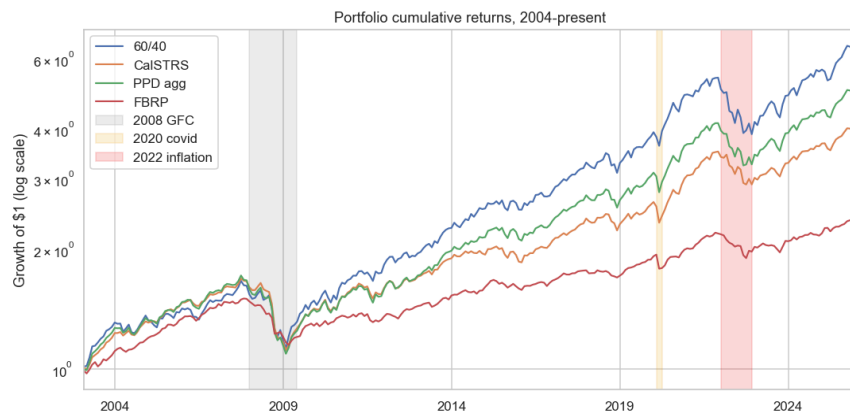
While the benchmark portfolios appear diversified across asset classes, their risk exposures remain heavily concentrated in equity, credit, and duration-related factors. The macro factor decomposition shows that the 60/40, CalSTRS, and PPD aggregate portfolios are dominated by Real Rate and Credit risk, with relatively little exposure to Growth and Inflation factors. The FBRP portfolios allocates risk more evenly across Real Rate and Credit exposures, reducing hidden macro-factor concentration.



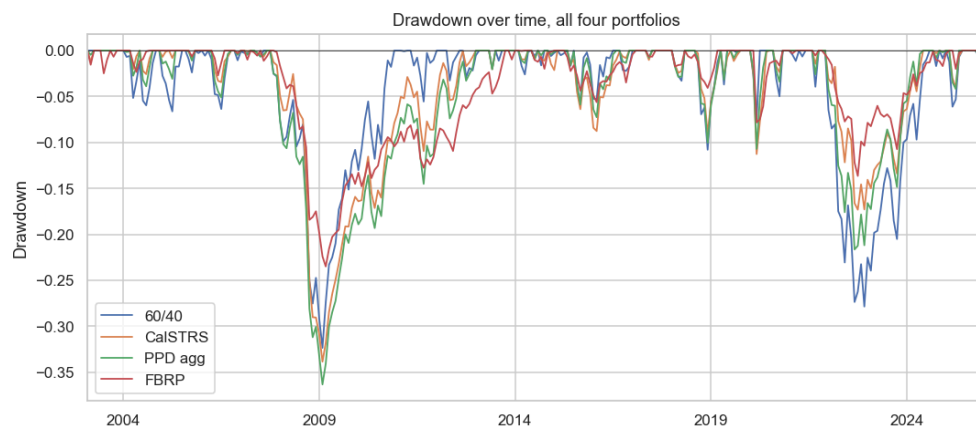
From 2004–2026 backtesting, FBRP has an annualized volatility 5.3%, much lower than benchmarks that are more than 10%. It also has much lower Conditional VaR 3.6%, while benchmarks are all greater than 5%. Although FBRP has a lower annualized return 3.9%, its Sharpe ratio still outperforms, with a much lower drawdown 23.5% compared with benchmarks that are more than 30%.

	Ann.Ret	Ann.Vol	Sharpe	Max DD	CVaR(5%)
60/40	0.084	0.106	0.792	-0.323	-0.067
CalSTRS	0.063	0.080	0.787	-0.338	-0.057
PPD agg	0.073	0.091	0.800	-0.363	-0.061
FBRP	0.039	0.053	0.729	-0.235	-0.036

The cumulative return results shows that although FBRP underperformed in total return, it experienced materially smaller losses during major stress periods, particularly during the 2008 Global Financial Crisis and the 2022 inflation shock, demonstrating stronger downside protection and portfolio resilience.



During stress periods, FBRP performs the best. In 2022, the benchmarks lost more than 17%, but FBRP lost only 9.95%, with significantly lower drawdown 12.83%. In 2008, FBRP also outperformed benchmarks on a much lower drawdown. In 2020, all portfolios equivalently suffered. Overall, the results show that diversification across macro risk factors provides stronger downside protection than traditional asset-class diversification.



6. Conclusion

The results show that traditional pension portfolios remain highly exposed to common macroeconomic risk despite appearing diversified across asset classes. Consistent with the project's original objective, the FBRP framework reallocates portfolio risk more evenly across macro factors. This leads to materially lower drawdown and tail risk, particularly during inflationary and rising-rate regimes such as 2022. Although the strategy sacrifices some return, it improves downside risk resilience and funding-ratio stability, which are critical objectives for long-horizon pension investors.

7: Caveats:

Several limitations should be noted. First, the FBRP optimization is conducted in-sample using the full 2004–2026 period, so the realized performance likely overstates what a real-time rolling implementation would achieve. Second, the Growth and Inflation contributions remain relatively small because the analysis relies on passive index proxies with limited sensitivity to these factors. Third, the real estate and private-equity proxies may understate true volatility due to appraisal smoothing and stale NAV effects, which could affect the measured diversification benefits.